

Wearable Robotics

LZO: Control of lower-limb powered prostheses and orthoses.

Prof. Bolívar

Nov. 13th, 2023

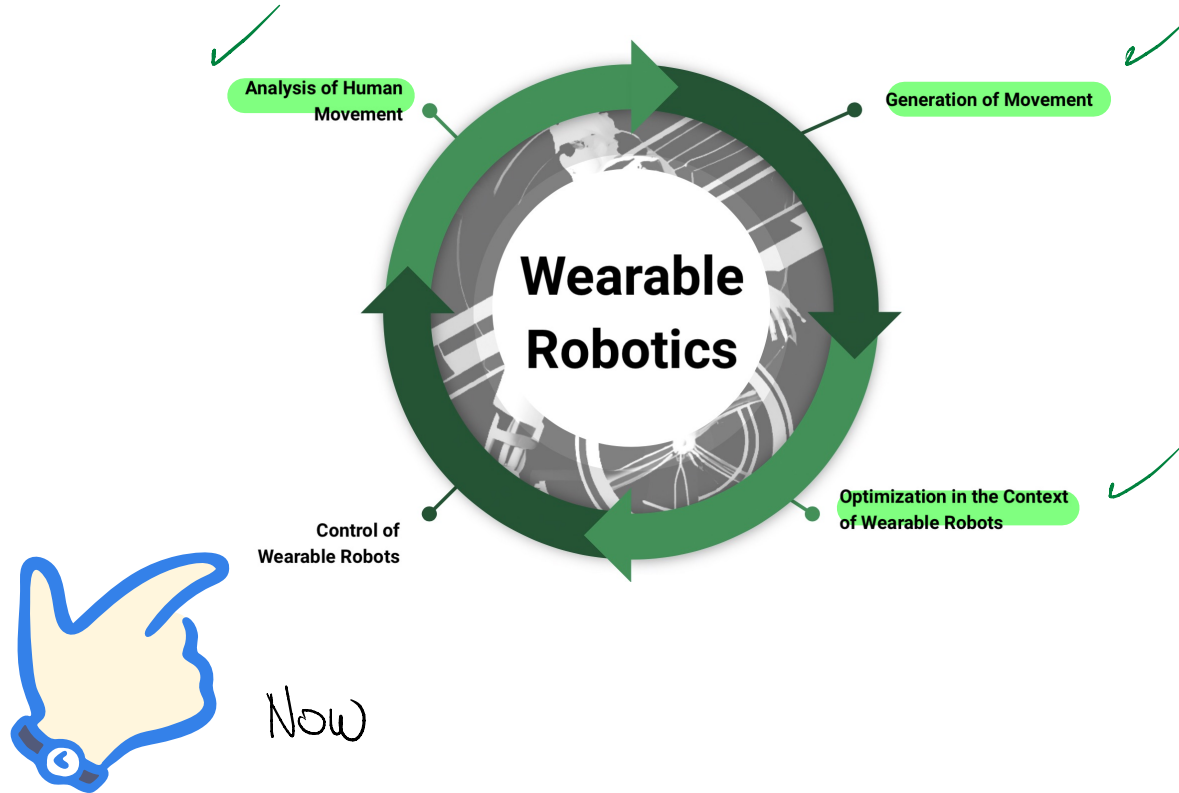
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Impedance Control: An Approach to Manipulation: Part I—Theory

Manipulation fundamentally requires the manipulator to be mechanically coupled to the object being manipulated; the manipulator must not be rigid or perfectly controlled. This kinematic paper presents an approach to the control of flexible structures having a manipulator and an environment. In Part I this approach is extended to controlling the mechanical or impedance behavior of the system. Control of position or force alone is inadequate; control of dynamic behavior is also required. It is shown that an impedance is a fundamentally nonlinear function of the system's dynamic response to the environment. It is shown that the environment provides useful signals; the manipulator must be an impedance. A consideration of a linear impedance approach is followed by a brief discussion of nonlinear impedance approaches. The control of force from the control of impedance while preserving the dynamic properties of the force network. It is shown that impedance is the appropriate measure of the mechanical or impedance even when they are nonlinear.



Control strategies for active lower extremity prosthetics and orthotics: a review
Taylor et al.



Powered Prostheses
&
Powered Orthoses

Why?

controllers
to

- Assist
- Challenge

Goal: Improve mobility
& quality of life

Framework for
controller design

Mid-level:
state machines
Phase variables

High-level:
Intention Estimation

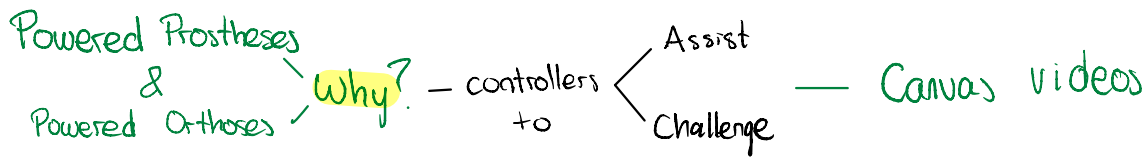
Similarities with
the Central Nervous System

Available signals for
control

- EMG
- Forces
- IMU
- BMI

Available joint-level
control strategies

- Position
- Admittance
- Impedance
- Torque
 - Time
 - Amplitude



Side effects of passive prostheses:

- Compensation from the Hip (Video)
- Increases the risk for osteoarthritis [4]
- Back pain [4]
- More than 60% of metabolic energy to walk than people without amputation [9][10]
- Walk around 40% slower
- The metabolic burden is higher for tasks that require mechanical energy from the knee and ankle (e.g., sit-stand, stairs) [11][12][13]
- Above-knee amputation increases the likelihood of falling. [14]

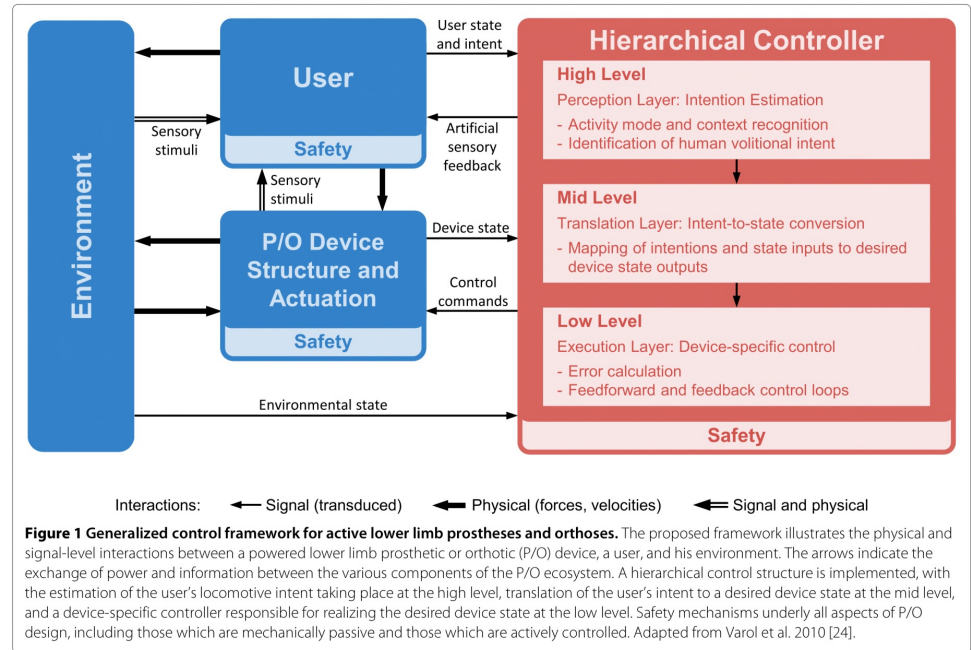
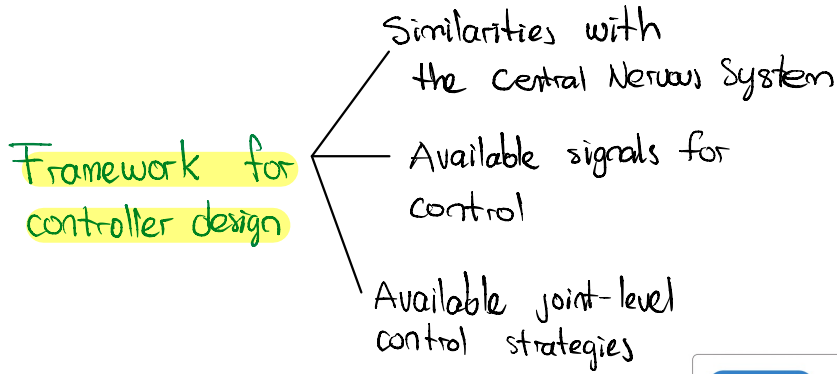
What kind of prosthesis is adequate? It depends on the mobility goals and activity level.

Define activity level (K-level from 0-4)

* What are K levels? (Guide from Ottobock)

- * Level 0 -- Does not have the ability or potential to ambulate or transfer safely with or without assistance and a prosthesis does not enhance their quality of life or mobility.
- * Level 1 -- Has the ability or potential to use a prosthesis for transfers or ambulation on level surfaces at fixed cadence. Typical of the limited and unlimited household ambulator.
- * Level 2 -- Has the ability or potential for ambulation with the ability to traverse low-level environmental barriers such as curbs, stairs, or uneven surfaces. Typical of the limited community ambulator.
- * Level 3 -- Has the ability or potential for ambulation with variable cadence. Typical of the community ambulator who has the ability to traverse most environmental barriers and may have vocational, therapeutic, or exercise activity that demands prosthetic utilization beyond simple locomotion.
- * Level 4 -- Has the ability or potential for prosthetic ambulation that exceeds basic ambulation skills, exhibiting high impact, stress, or energy levels. Typical of the prosthetic demands of the child, active adult, or athlete.

- [9] R. S. Gailley et al., "Unilateral lower-limb loss: Prosthetic device use and functional outcomes in servicemembers from Vietnam war and OIF/OEF conflicts," J. Rehabil. Res. Dev., vol. 47, no. 4, p. 317, 2010.
- [10] B. J. Hafner, J. E. Sanders, J. Czerniecki, and J. Ferguson, "Energy storage and return prostheses: does patient perception correlate with biomechanical analysis?," Clin. Biomech., vol. 17, no. 5, pp. 325-344, 2002.
- [4] H. Liu, C. Chen, M. Hanson, R. Chaturvedi, S. Matke, and R. Hillestad, Economic Value of Advanced Transfemoral Prosthetics. Rand Corporation, 2017.
- [11] F. Sup, H. A. Varol, and M. Goldfarb, "Upslope walking with a powered knee and ankle prosthesis: Initial results with an amputee subject," IEEE Trans Neural Sys Rehab Eng, vol. 19, no. 1, pp. 71-78, 2011.
- [12] B. Lawson, H. Varol, A. Huff, E. Erdemir, and M. Goldfarb, "Control of stair ascent and descent with a powered transfemoral prosthesis," IEEE Trans. Neural Sys. Rehab. Eng., vol. 21, no. 3, pp. 466-473, 2013.
- [13] A. Shultz, B. Lawson, and M. Goldfarb, "Running with a powered knee and ankle prosthesis," IEEE Trans. Neural Sys. Rehab. Eng., vol. 23, no. 3, pp. 403-412, 2015.
- [14] J. Kulkarni, C. Toole, R. Hiron, S. Wright, and J. Morris, "Falls in patients with lower limb amputations: Prevalence and contributing factors," Physiotherapy, vol. 82, no. 2, pp. 130-136, 1996



Framework for controller design

Similarities with the Central Nervous System

Available signals for control

Available joint-level control strategies

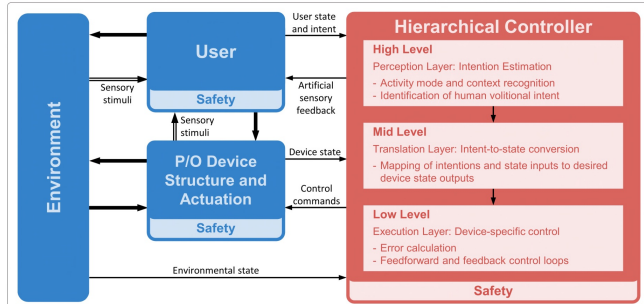


Figure 1 Generalized control framework for active lower limb prostheses and orthoses. The proposed framework illustrates the physical and signal-level interactions between a powered lower limb prosthetic or orthotic (P/O) device, a user, and his environment. The arrows indicate the exchange of power and information between the various components of the P/O ecosystem. A hierarchical control structure is implemented, with the estimation of the user's locomotive intent taking place at the high level, translation of the user's intent to a desired device state at the mid level, and a device-specific controller responsible for realizing the desired device state at the low level. Safety mechanisms underlie all aspects of P/O design, including those which are mechanically passive and those which are actively controlled. Adapted from Varol et al. 2010 [24].

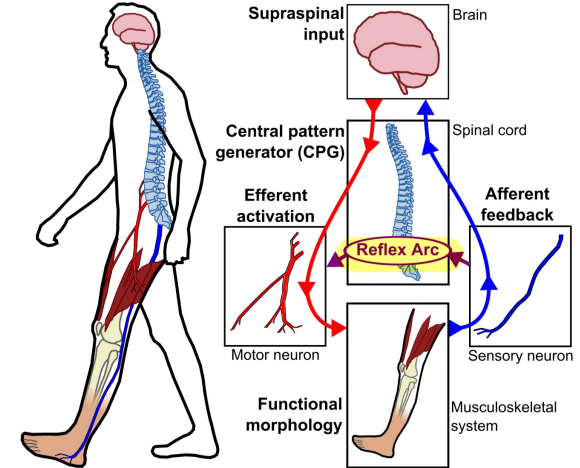
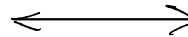
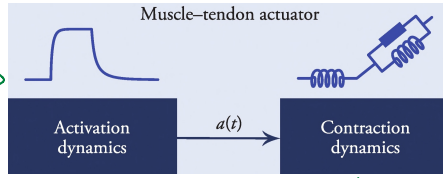


Figure 2 Nominal sensory-motor control loop for human locomotion. Motion intentions originate from supraspinal input, which along with afferent feedback serves to modulate basic underlying locomotor patterns within a network of spinal interneurons, commonly referred to as the central pattern generator (CPG). Efferent stimulation is transmitted through motor neurons to individual muscle groups, which are recruited to effect the movement. Afferent feedback, including that from proprioceptors of the muscles and joints and mechanoreceptors of the skin, is used to directly modulate motor commands via mono- and polysynaptic reflex arcs, thus contributing to the efficiency of gait under normal conditions and stability of gait in the face of unexpected perturbations. Sensory information is also transmitted to the brain, where it is combined with higher level inputs from the visual, auditory, and vestibular systems to provide information required for the maintenance of balance, orientation and control of precise movements.

Neural Signal

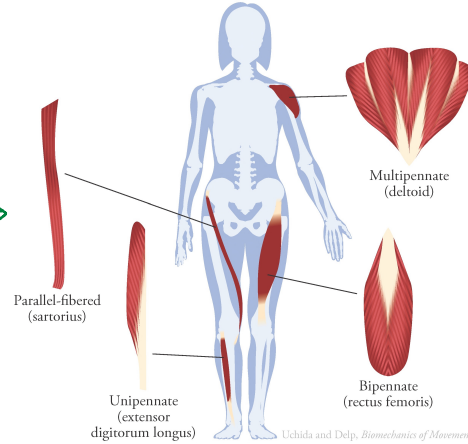
Motor
Efferents



lectures 6-7

Musculoskeletal Geometry

Forces



Central
Nervous
System



Proprioception

Biological Sensors
in skin, muscles,
tendons

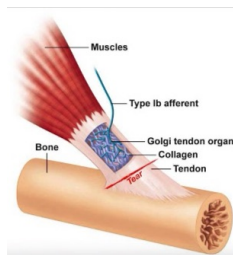
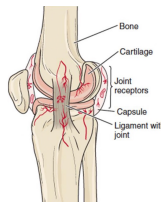
Body Dynamics

Motor
Afferents

Abstract

In the past, the peripheral sense organs responsible for generating human position sense were thought to be the slowly adapting receptors in joints. More recently, our views have changed and the principal position sensor is now believed to be the muscle spindle. Joint receptors have been relegated to the lesser role of acting as limit detectors when movement approaches the anatomical limit of a joint. In a recent experiment concerned with position sense at the elbow joint, measured in a pointing task over a range of forearm angles, we have observed falls in position errors as the forearm was moved closer to the limit of extension. We considered the possibility that as the arm approached full extension, a population of joint receptors became engaged and that they were responsible for the changes in position errors. Muscle vibration selectively engages signals of muscle spindles. Vibration of elbow muscles undergoing stretch has been reported to lead to perception of elbow angles beyond the anatomical limit of the joint. The result suggests that spindles, by themselves, cannot signal the limit of joint movement. We hypothesize that over the portion of the elbow angle range where joint receptors become active, their signals are combined with those of spindles to produce a composite that contains joint limit information. As the arm is extended, the growing influence of the joint receptor signal is evidenced by the fall in position errors.

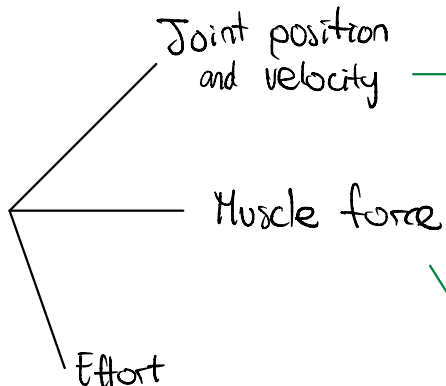
Keywords Muscle spindle · Thixotropy · Proprioception · Position sense · Joint receptor



Proprioception

Sense that let us perceive the location and action of parts in our body.

Signals arise from sensory receptors in the muscle, skin, and joints.

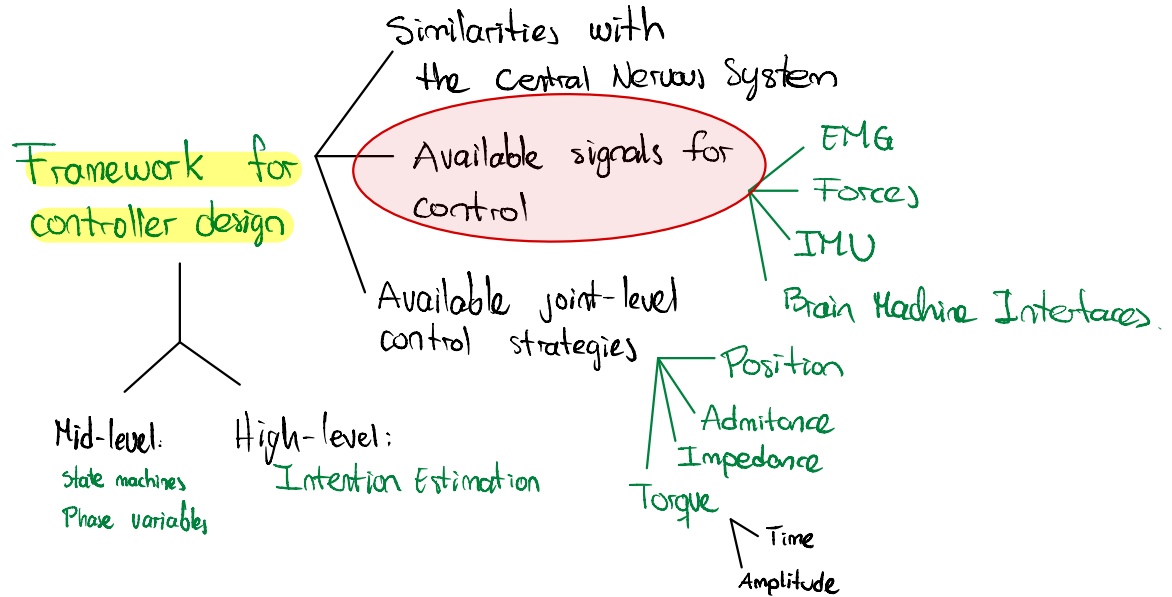


Joint Receptors

- Mechanoreceptors in/around joints and skin
- Provides information about ROM limits

Golgi Tendon Organ

- located in capsule between muscle body and tendon
- Mostly sensitive to active tension, rather than active stretch



Example of semi-raw EMG Data

Data Descriptor | [Open access](#) | Published: 06 July 2023

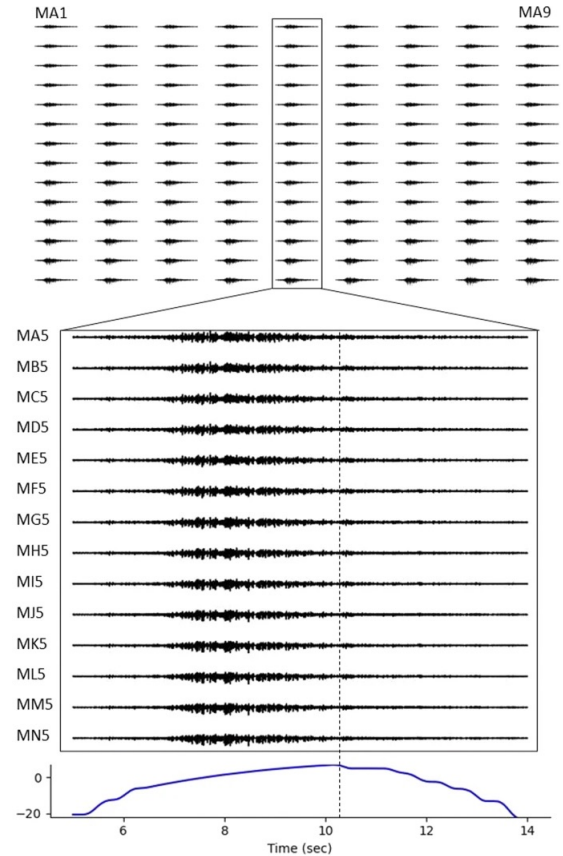
High-density Surface and Intramuscular EMG Data from the Tibialis Anterior During Dynamic Contractions

[J. Courtney Bradford](#) , [Andrew Tweedell](#) & [Logan Leahy](#)

[Scientific Data](#) **10**, Article number: 434 (2023) | [Cite this article](#)

1394 Accesses | 5 Altmetric | [Metrics](#)

Minimally processed HDsEMG data from a representative isotonic trial. Data were band pass filtered 20–400 Hz for visualization purposes. HDsEMG traces are plotted for the full grid of 126 channels. For more detail, we zoomed in for a single column of 14 channels plotted along with sagittal plane, ankle joint angle. The vertical dashed line roughly denotes the transition to eccentric contraction.



How can we get useful information from this data?

Electro Myo Graphy

Measurement of muscle action potential

Surface EMG

indwelling/intramuscular EMG



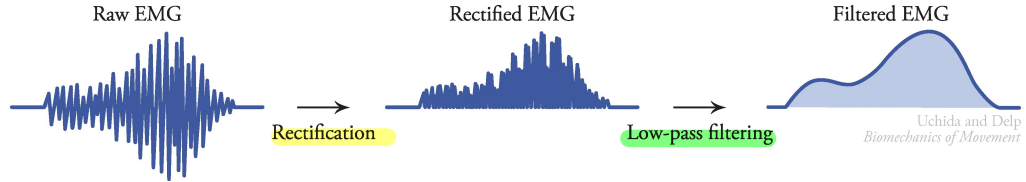
DELSYS SENS

EMG includes electrical stimulation due to orderly recruitment and rate encoding

To be aware,

EMG is sensitive to:

- Motion artifacts
- Changes in skin impedance
- Muscle fatigue
- Changes in motor recruitment patterns



- ① High-Pass EMG signal to remove drift
- ② Rectify EMG $\rightarrow$ Rectified EMG = $| \text{Low-Pass EMG} |$
- ③ Low-Pass EMG $\rightarrow$ Obtain the envelope.
RMS moving window technique.
- ④ Normalize $\rightarrow$ Maximum Voluntary Contraction
Relaxed state
- ⑤ Get excitation signal $ue[0, 1]$

Forces/Moments (effort)

Direct measurements of effort
require forces/moments to flow
through the sensor

→ load cell
→ Insole

Well-posed problem in prosthetic
applications because loads pass
directly through the prosthesis

Non-trivial problem in orthoses/exoskeletons
as the loads transfer partially through
the device.

Available
"sensors"

Insoles



6-axis
Load Cells



Forward Dynamics

From kinematics to
joint kinetics

Actuator torque* — Depending on the transmission
and electronic driver.

Insoles



Pros

- Portable
- Embedded IMUs
- Reasonable accuracy

Cons

- May not provide shear forces, which are important for complete GFC
- A pair could be in the order of \$1k.

6-axis
load cell



Pro

Complete wrench
information (3 forces + 3 moments!)

can be low profile and low weight



SRI Sunrise Instruments

PRODUCTS APPLICATIONS SUPPORT NEWS ABOUT US CONTACT US

English

Model	Description	Measuring Range(N/m)				Dimension (mm)			Weight (Kg)	
		FX, FY	FZ	MX, MY	MZ	OD	Height	ID		
M3552B	EXTRA THIN 6 AXIS LC	150	250	2.25	2.25	30	9.2	5	0.01	Download
M3552C	EXTRA THIN 6 AXIS LC	300	500	4.5	4.5	30	9.2	5	0.03	Download
M3552D	EXTRA THIN 6 AXIS LC	600	1000	9	9	30	9.2	5	0.03	Download
M3553B	EXTRA THIN 6 AXIS LC	150	250	3.5	3.5	45	9.2	9	0.03	Download
M3553C	EXTRA THIN 6 AXIS LC	300	500	7	7	45	9.2	10	0.06	Download
M3553D	EXTRA THIN 6 AXIS LC	600	1000	13.5	13.5	45	9.2	10	0.06	Download
M3553E	EXTRA THIN 6 AXIS LC	1200	2000	27	27	45	9.2	10	0.06	Download
M3554C	EXTRA THIN 6 AXIS LC	300	500	10	10	60	9.2	21	0.11	Download
M3554D	EXTRA THIN 6 AXIS LC	600	1000	20	20	60	9.2	21	0.11	Download
M3554E	EXTRA THIN 6 AXIS LC	1200	2000	40	40	60	9.2	21	0.11	Download
M3555D	EXTRA THIN 6 AXIS LC	600	1000	40	40	90	9.2	45	0.26	Download
M3564C	EXTRA THIN 6 AXIS LC	1200	1200	40	30	60	10	7	0.06	Download
M3564F	EXTRA THIN 6 AXIS LC	2500	5000	200	100	65	9.2	10	0.19	Download

Cons

require careful signal amplification

About \$3k each

Inertial Measurement Unit



It is a combination
of multiple sensors
typically

check MATLAB mobile → sensors →

Acceleration		☒
10 Hz	-0.027	
X (m/s²)	-0.009	
Y (m/s²)	0.001	
Z (m/s²)	9.801	

Accelerometer

- Measures linear accelerations in each of the three axes rigidly attached to the sensor

Gyroscope

- Measures angular velocities in each of the axes

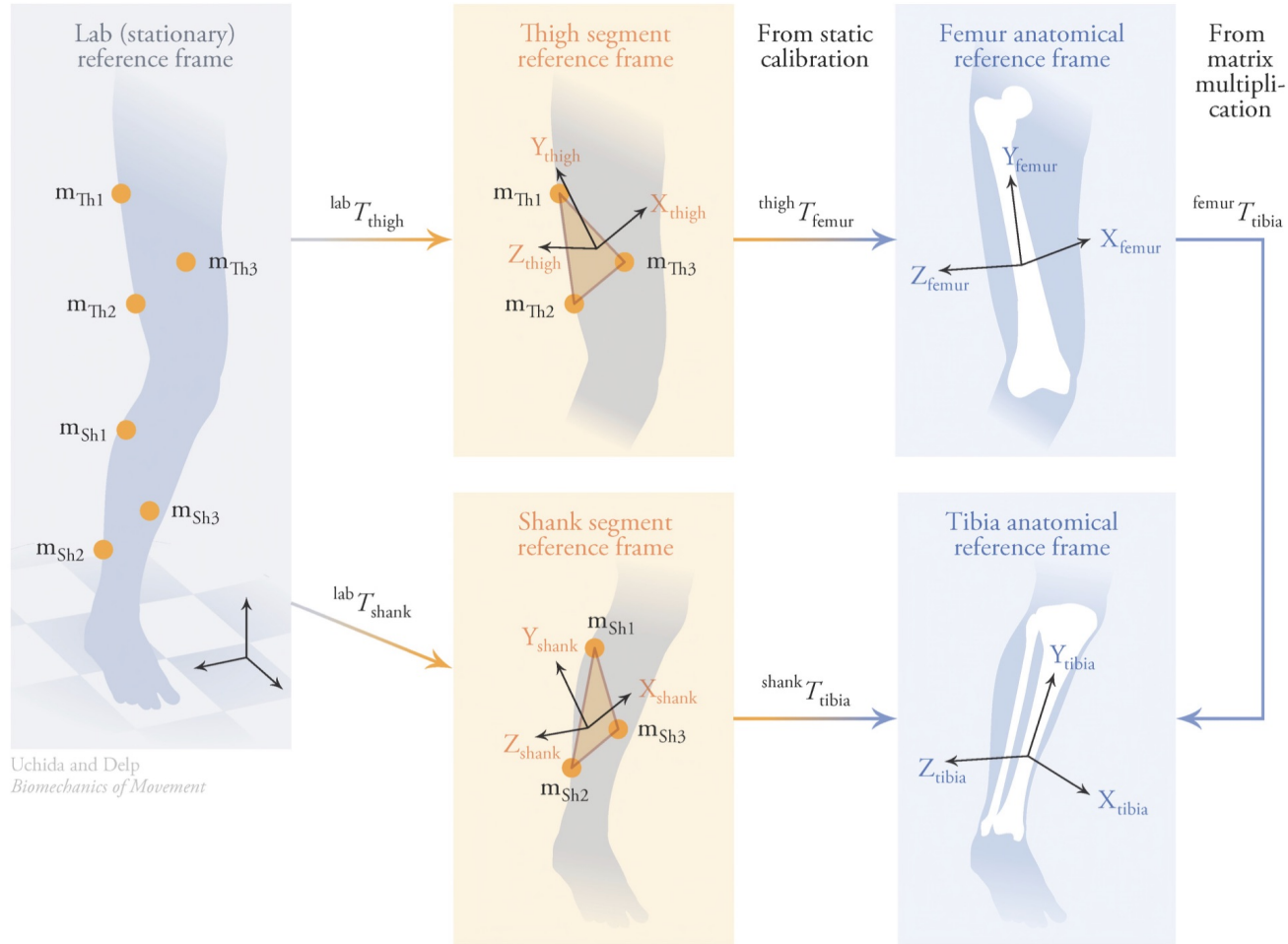
Magnetometer (Optional)

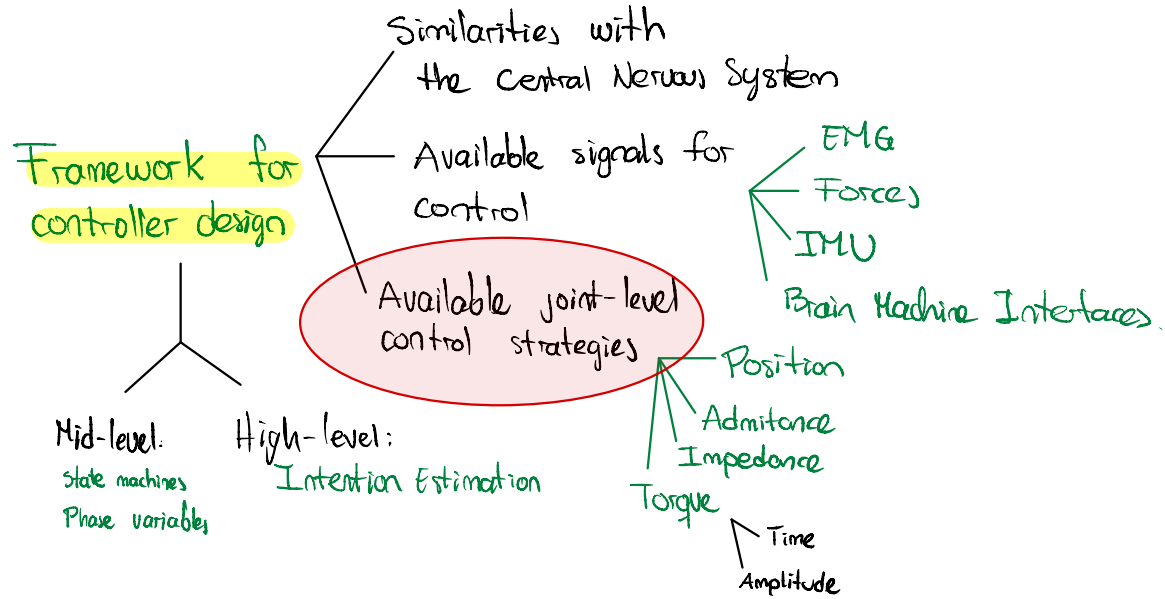
Strength of a magnetic field.

Note: IMUs can report orientation relative to a global frame.

This is likely an estimation from a kalman filter not a direct measurement!

IMUs can
provide segment
reference frames
to estimate
joint angles!

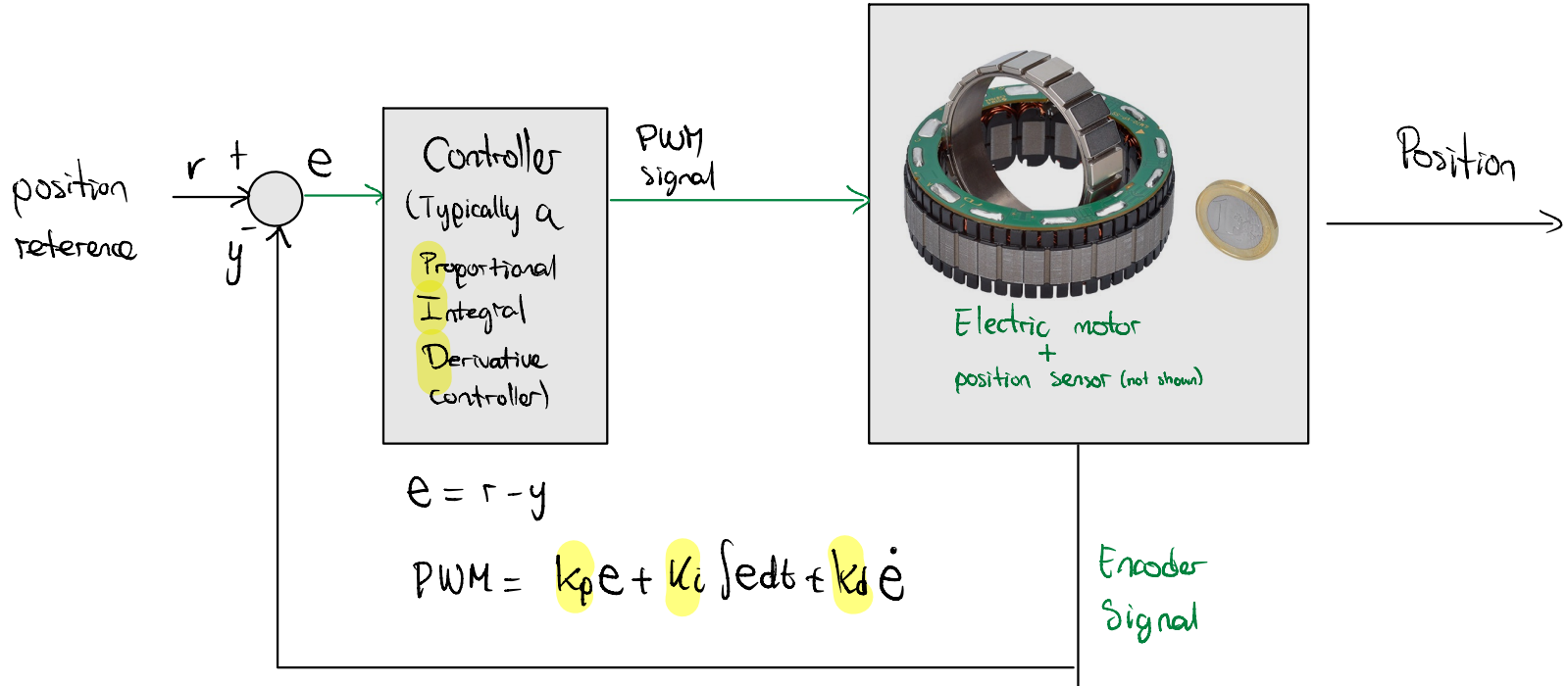




Position Control

Well-documented control problem

It is not ideal for human-robot interaction

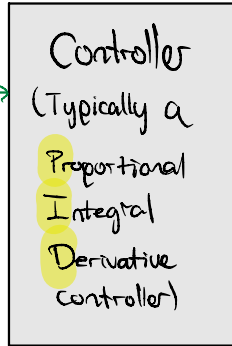
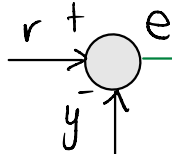


Torque Control

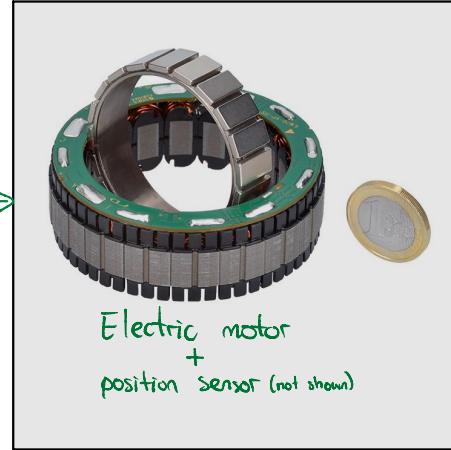
Well-documented control problem
The quality of torque tracking depends on the transmission and associated torque "losses"

— Torque control enables dynamic motion that connects with the equations of motion "F=ma" approach.

Torque reference



PWM signal



Torque

$$e = r - y$$

$$\text{PWM} = k_p e + k_i \int e dt + k_d \dot{e}$$

Torque signal

Dynamic interaction between two physical systems requires that one must complement each other:

If one is an admittance the other is an impedance.

Physical equivalence.

$$\tau_m = -b_m \dot{q}_m$$

or

$$\tau_m = k_p q_m$$

Impedance

Control

Causality

Instantaneous power flow is always defined as the product of two variables

Control framework when the mechanical work by the robotic joint is not negligible

$$dw = F \cdot dx$$

Instantaneous Power Flow = effort · flow

↓
e.g. force,
Voltage

↓
e.g. velocity
and current

Important physical constraint: No system can determine those two variables

Definition: Impedances accept flow (e.g. motion) inputs and yield effort (e.g. forces)

